

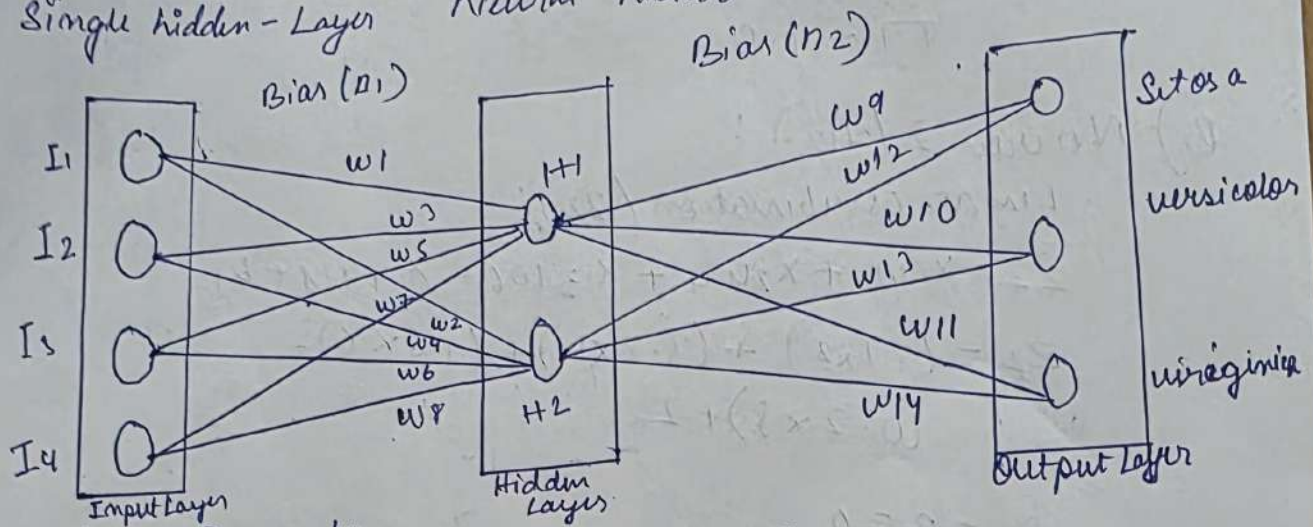
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Mathematical Calculation

Single hidden - Layer Neural Network



1. Given Parameter

Input vector: $x = [5.1, 3.2, 1.4, 0.2]$

Hidden weights (H1): $w_1 = 1, w_3 = 3$
 $w_5 = 6, w_7 = 8$

Output Hidden weights (H2): $w_2 = 2, w_4 = 4, w_6 = 6, w_8 = 8$

output weights (from H1): $w_9 = 1.3$
 $w_{10} = 0.5, w_{11} = 1.2$

output weights (from H2): $w_{12} = 0.5, w_{13} = 0.2, w_{14} = 0.4$

Bias: $b_1 = 2, b_2 = 0.1$

2. Hidden Layer calculation

A) Neuron (H1):

Linear combination ($\neq 1$):

$$Z_1 = x_1 w_1 + x_2 w_3 + x_3 w_5 + x_4 w_7 + b_1$$

$$Z_1 = (5.1 \times 1) + (3.2 \times 3) + (1.4 \times 6) + (0.2 \times 8) + 2$$

$$Z_1 = 5.1 + 9.6 + 8.4 + 1.6 + 2$$

$$Z_1 = 26.7$$

Activation (σ_1):

$$\sigma_1 = \frac{1}{1+e^{-z_1}} = \frac{1}{1+e^{-25.1}}$$

$$\sigma_1 \approx 1.0$$

B) Neuron 2 (H_2):

Linear combination (z_2):

$$z_2 = x_1 w_2 + x_2 w_4 + x_3 w_6 + x_4 w_8 + b_1$$

$$z_2 = (5.1 \times 2) + (3.2 \times 4) + (1.6 \times 6) + (0.2 \times 8) + 2$$

$$z_2 = 35.0$$

Activation (σ_2):

$$\sigma_2 = \frac{1}{1+e^{-z_2}} = \frac{1}{1+e^{-35.0}}$$

$$\sigma_2 \approx 1.0$$

3) Output Layer calculation
(combining both neurons)

Al Sitor a ($z_{sitor a}$):

$$z_{sitor a} = (\sigma_1 \times w_9) + (\sigma_2 \times w_{12}) + b_2$$

$$= (1.0 \times 1.3) + (1.0 \times 0.5) + 0.1$$

$$= 1.3 + 0.5 + 0.1$$

$$z_{sitor a} = 1.9$$

b) Versicolor ($z_{\text{versicolor}}$):

$$\begin{aligned} z_{\text{versicolor}} &= (\sigma_1 \times w_{10}) + (\sigma_2 \times w_{11}) + b_L \\ &= (1.0 \times 0.5) + (1.0 \times 0.2) + 0.1 \\ &= 0.5 + 0.2 + 0.1 \end{aligned}$$

$$z_{\text{versicolor}} = 0.8$$

c) virginica ($z_{\text{virginica}}$):

$$\begin{aligned} z_{\text{virginica}} &= (\sigma_1 \times w_{11}) + (\sigma_2 \times w_{14}) + b_L \\ &= (1.0 \times 1.2) + (1.0 \times 0.4) + 0.1 \\ &= 1.2 + 0.4 + 0.1 \end{aligned}$$

$$z_{\text{virginica}} = 1.7$$

4) Final Prediction

• output vector: $(1.9, 0.8, 1.7)$

• Prediction: Setosa [Highest Sum 1.9]